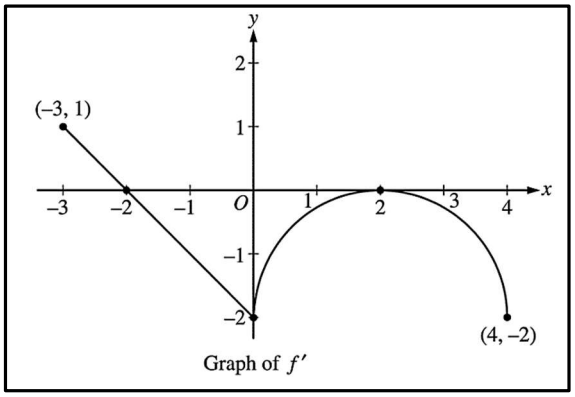
Worksheet 6-4 & 6-5 C
FTC Part 2 and Graphical Analysis #3
AP Calculus

Name_____

1.) **2003 AB 4**

Let f be a function defined on the closed interval $-3 \le x \le 4$ with $f(0) = 3$. The graph of f', the derivative of f, consists of one line segment and a semicircle, as shown at the right.



Graph of f'

(a) On what intervals, if any, is f increasing? Justify your answer.

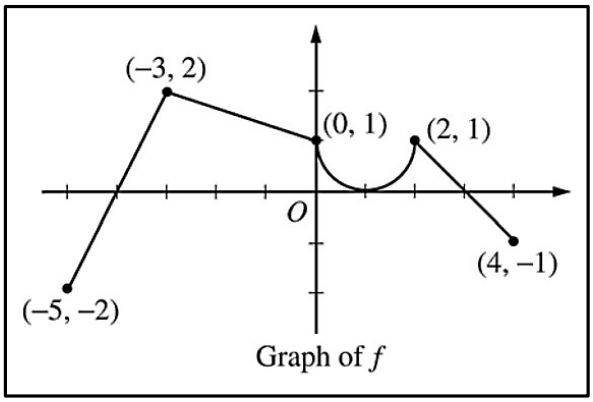
(b) Find the x coordinate of each point of inflection of the graph of f on the open interval $-3 < x < 4$. Justify your answer.

(c) Find an equation for the line tangent to the graph of f at the point $(0, 3)$.

(d) Find $f(-3)$ and $f(4)$. Show the work that leads to your answers.

2.) **2004 AB5**

The graph of the function f shown at the right consists of a semicircle and three line segments. Let g be the function given by $g(x) = \int_{-3}^{x} f(t)dt$



Graph of f

(a) Find $g(0)$ and $g'(0)$

(b) Find all values of x in the open interval $(-5, 4)$ at which g attains a relative maximum. Justify your answer.

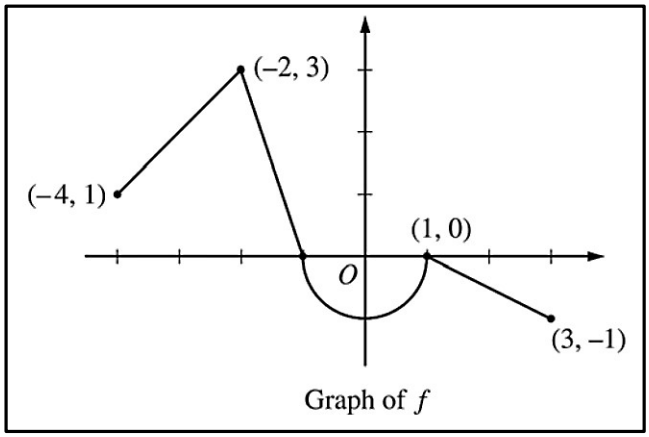
(c) Find the absolute minimum value of g on the closed interval $[-5, 4]$. Justify your answer.

(d) Find all values of x in the open interval $(-5, 4)$ at which the graph of g has a point of inflection.

3.) **2012 AB3**

Let f be the continuous function defined on $[-4, 3]$ whose graph, consisting of three line segments and a semicircle centered at the origin, is given at the right.

Let g be the function given by $g(x) = \int_1^x f(t)dt$



Graph of f

(a) Find the values of $g(2)$ and $g(-2)$.

(b) For each of $g'(-3)$ and $g''(-3)$, find the value or state that it does not exist.

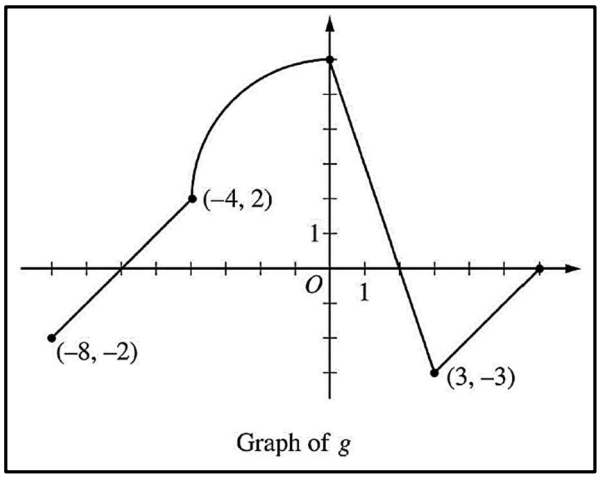
(c) Find the x coordinate of each point at which the graph of g has a horizontal tangent line. For each of these points, determine whether g has a relative minimum, relative maximum, or neither a minimum nor maximum at the point. Justify your answers.

(d) For $-4 < x < 3$, find all values of x for which the graph of g has a point of inflection. Explain your reasoning.

4.) **2016 PE AB3**

A continuous function g is defined on the closed interval $-8 \le x \le 6$. The graph of g, shown above, consists of three line segments and a quarter circle centered at the point $(0, 2)$. Let f be the function given by

$$f(x) = \int_{-8}^{x} g(t)dt.$$



Graph of g

(a) Find all values of x in the interval $-8 < x < 6$ at which f has a critical point. Classify each critical point as the location of a local minimum, a local maximum, or neither. Justify your answers.

(b) Find $f(0)$

(c) Find $\lim_{x \to -4} \frac{f(x)}{x^2 + 4x}$

(d) Let h be the function defined by $h(x) = \frac{g(x)}{x^2 + 1}$. Find $h'(1)$.